How Rockets Work

Newton’s laws, four forces, four systems and two kinds of engine, with ISRO’s LVM3 as the worked example.

Isagi · November 2026 · Theory from NASA Glenn’s Beginner’s Guide to Rockets; ISRO figures checked 9 November 2026

HOW TO USE THIS CHAPTER

- **Print it single-sided** and write in the margins. Read one section at a time.
- After each section, **close the page and write three lines** in your own words. If you can’t, read it again.
- In the worked examples, **cover the answer**, do the sum with a calculator, then check. Use g = 9.81 m/s².
- The practice problems and glossary come after the summary, and the answer key is on the last page.

By the end of this chapter you can: explain why a rocket works in empty space; name the four forces and the four systems of a rocket; and compute the lift-off weight, propellant share, burn rate and thrust-to-weight ratio of a real launch vehicle by hand.

1 Why a rocket moves

Every rocket runs on one idea: **throw mass backwards and you get pushed forwards.** NASA Glenn states Newton’s three laws of motion this way. A body keeps its motion unless an unbalanced force acts on it (inertia). Acceleration depends on the force and the mass, **F = m × a**. And every action has an equal and opposite reaction.

The third law is the engine. The rocket burns propellant, the hot gas is flung out of the back, and the gas pushes the rocket the other way with an equal force. NASA calls that gas the **working fluid**. A jet engine or a propeller uses the surrounding air as its working fluid, so it stops working where there is no air. A rocket carries its fuel *and* its oxygen

source (the **oxidiser**) and uses its own exhaust, so it works just as well in a vacuum. That is the biggest difference between a rocket and an aircraft.

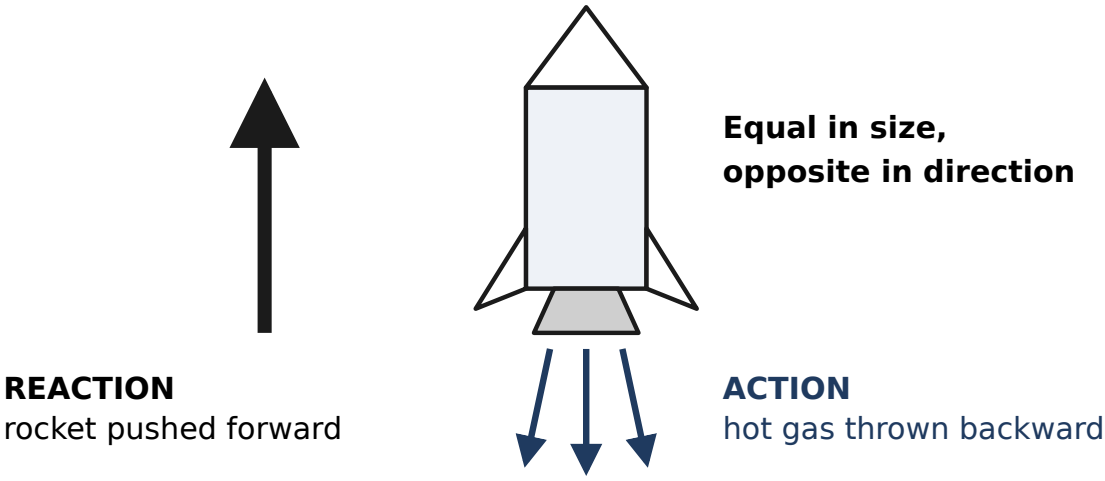


Figure 1. Newton’s third law in a rocket. The engine throws gas down (action); the gas pushes the rocket up (reaction).

The second law tells you what happens next. If the upward push (**thrust**) is bigger than the weight, the net force points up and the rocket accelerates. One warning from NASA: F = m × a holds only for constant mass. A rocket burns away most of its mass, so the full statement is about momentum, and we will build it properly in Chapter 2. For now, treat the mass as constant over a short moment. The first law explains the coast: once the engines stop, the rocket keeps moving, bent only by gravity and drag.

> **ISRO LENS**
> LVM3 injects its satellite into a 180 × 36,000 km transfer orbit about 974 seconds after lift-off. After that no engine is pushing the satellite. The first law and gravity carry it the rest of the way.

2 The four forces

In flight a rocket feels four forces: weight, thrust, drag and lift. Each is a **vector**, with a size and a direction, so you add them with direction in mind. The sum is the **net force**, and Newton’s second law turns that into motion.

- **Weight (W = m × g)** pulls towards the Earth's centre and acts through the **centre of gravity (CG)**.
- **Thrust (T)** is the engine's push, normally along the rocket's axis. Some rockets swivel the nozzle (**gimbal** it) so the thrust points slightly off the CG line. That makes a turning force and steers the rocket.
- **Drag (D)** is air resistance along the flight path. It depends on shape, size, speed and air density, and acts through the **centre of pressure (CP)**.
- **Lift (L)** on a rocket is not the lift of an aircraft. It is the aerodynamic *side* force from the fins and body, which helps keep the rocket pointing the right way and lets it be steered.

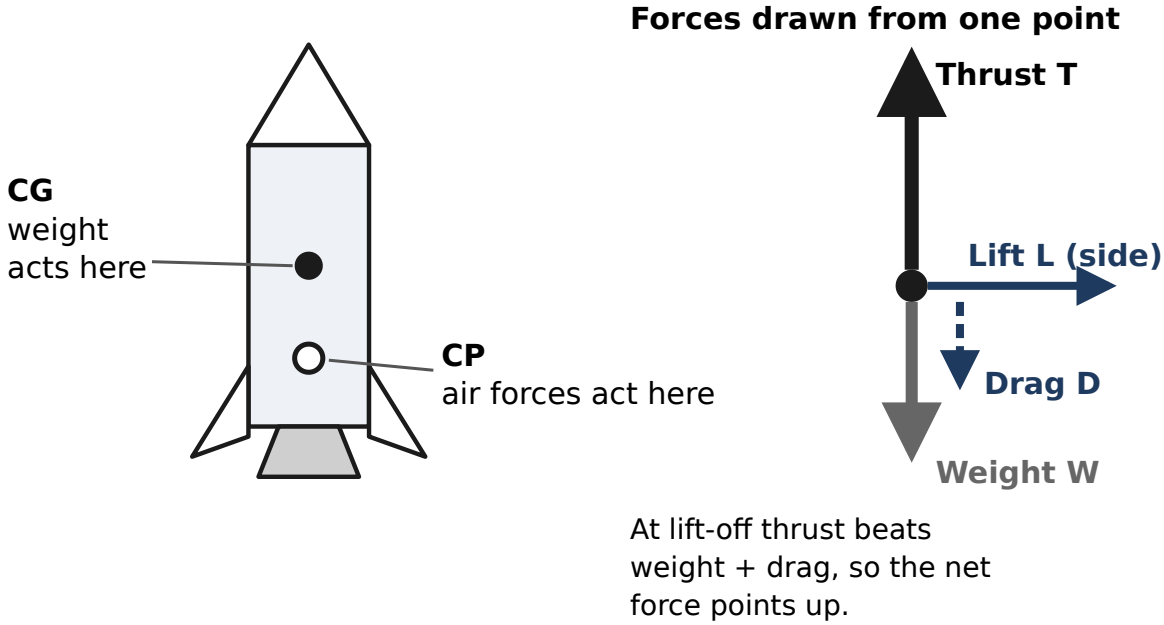


Figure 2. Left: where the forces act. CP sits behind (below) CG, which keeps the rocket stable. Right: the four forces as arrows. Arrow lengths show a rocket just after lift-off. Chapter 5 explains why CP must stay behind CG.

NASA lists four differences from an aircraft. **First**, an aircraft uses lift to beat weight; a rocket uses thrust. **Second**, an aircraft makes most of its air force with wings and tail; a rocket makes it with fins, nose cone and body. **Third**, an aircraft has far more lift than drag; a rocket usually has far more drag than lift. **Fourth**, an aircraft's forces stay roughly steady in cruise; a rocket's forces change dramatically during a flight, because it burns most of its mass, speeds up and climbs into thinner air.

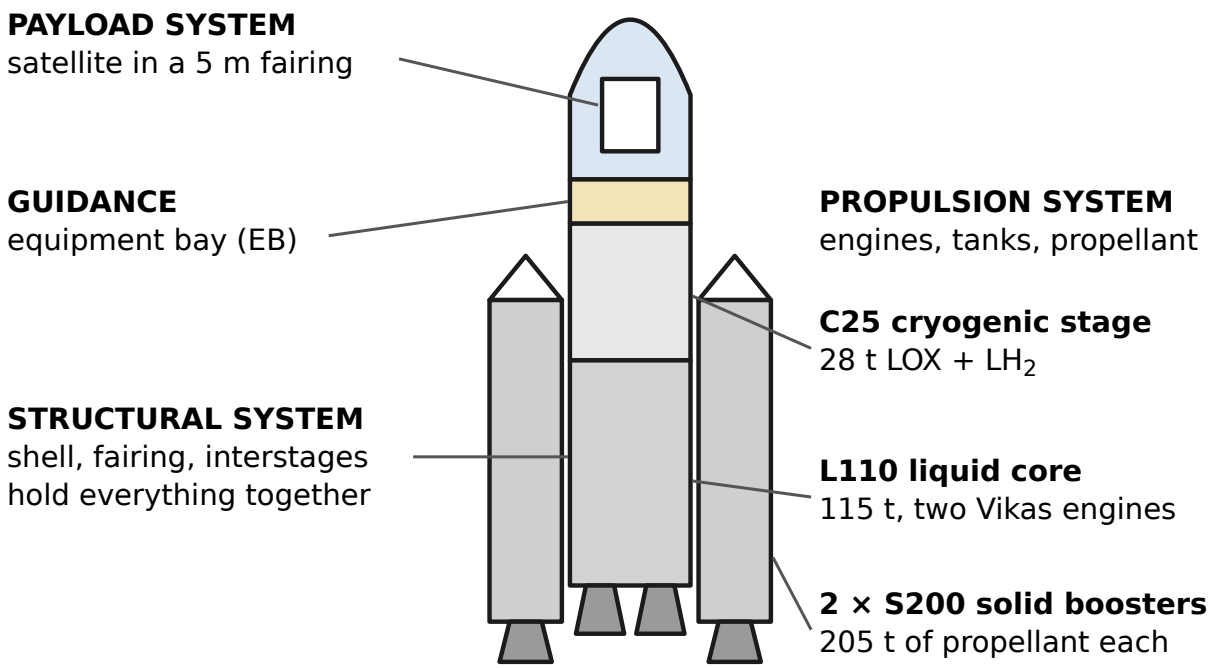
The one number to know: thrust-to-weight

The **thrust-to-weight ratio, T/W**, compares the push to the weight. If T/W is below 1, the rocket cannot leave the ground. Above 1, it accelerates upward. Ignoring drag, Newton's second law gives:

```
a = (T − W) / m = (T/W − 1) × g
Example: T/W = 1.5  →  a = 0.5 × 9.81 ≈ 4.9 m/s²
```

3 The four systems

A launch vehicle has thousands of parts. Engineers group them by job into four systems:



LVM3-style stack, hand-drawn, not to scale. Shading only separates parts; labels carry the meaning.

Figure 3. The four systems on an LVM3-style vehicle. ISRO's "encapsulated assembly" is the satellite plus its adaptor and fairing.

- **Structural system.** The frame: body, fairings and fins. It carries the loads of flight, keeps drag low, and must be strong yet light. Parts of it are dropped along the way (staging, Section 5).
- **Payload system.** Whatever the mission carries: satellites, instruments or people. It is the reason the rocket exists.

- **Guidance system.** Sensors, computers and communications that keep the rocket stable and steer it. The old German V-2 used vanes in the exhaust to deflect the thrust. Modern rockets mostly swivel the nozzle instead.
- **Propulsion system.** Engines, tanks, pumps, propellants and nozzle. In a full-size rocket this is most of what you see, because propellant is most of the mass.

For performance sums, engineers often lump everything except the propellant into one number, the **empty weight**. We use that idea in Chapter 2.

> **ISRO LENS**
> ISRO describes LVM3 as three stages plus the equipment bay (EB) and the encapsulated assembly (EA). The vehicle is 43.5 m tall with a 4 m body and a 5 m fairing. It lifts off at 640 tonnes and can place 8 t in low Earth orbit or 4 t in a geosynchronous transfer orbit.

4 Two kinds of engine

Rocket engines fall into two families. In a **liquid** engine, the fuel and oxidiser sit in separate tanks, and pumps feed them into a combustion chamber. In a **solid** engine, fuel and oxidiser are already mixed and cast into one block, the **grain**. An igniter lights it, and it burns until it is used up. In both cases the hot gas leaves through a **nozzle**, which turns heat into fast-moving exhaust.

> **ISRO LENS**
> LVM3 uses each type where it fits. **Solid** boosters give the huge, simple push needed at lift-off. A **liquid** core stage takes over for the long middle of the climb. The upper stage is **cryogenic**: it burns liquid oxygen and liquid hydrogen, which must be kept extremely cold. PSLV alternates solid, liquid, solid, liquid across its four stages. Chapter 1 shows why the cryogenic stage is the efficient one.

	Solid	Liquid
Propellant	Fuel and oxidiser mixed and cast into one block	Stored separately as liquids and pumped to the chamber
Start and stop	Once lit it burns until the propellant is gone; it cannot be switched off	Thrust stops when the propellant flow is shut off
Complexity	Simple: few moving parts	Heavier and more complex because of pumps and tanks
Handling	Easy to handle; can sit for years before firing	Propellants are loaded just before launch
ISRO example	S200 booster on LVM3; first stage of PSLV	Vikas engines in LVM3's L110 core; CE-20 in the C25 upper stage

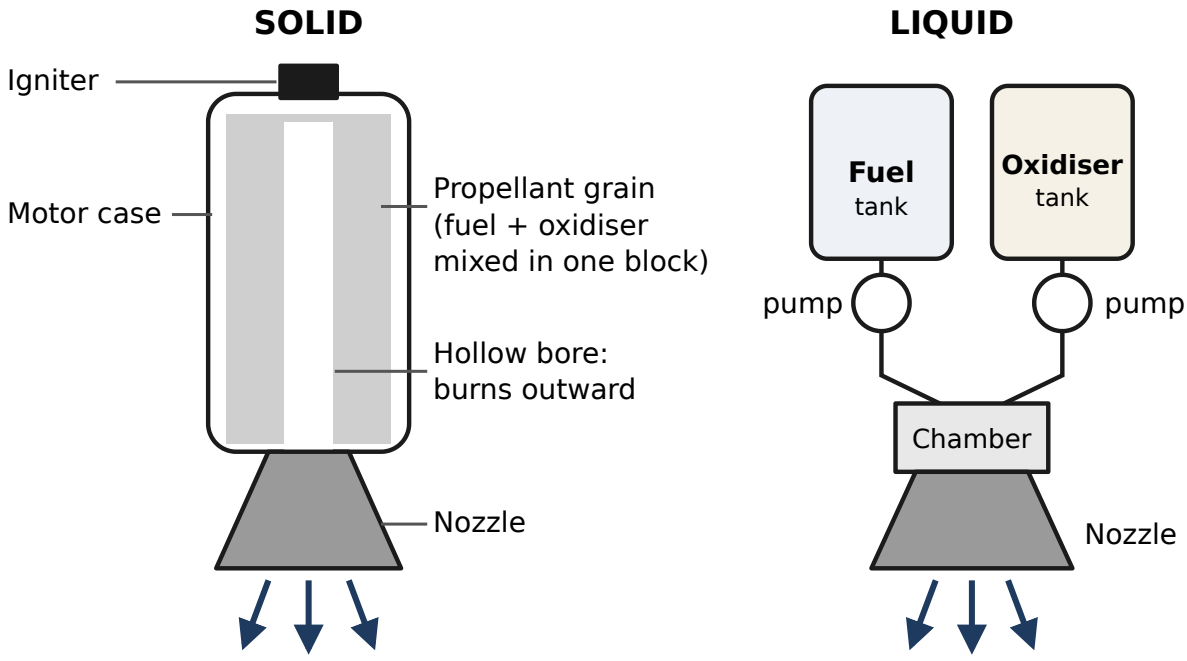


Figure 4. A solid motor is one cast block with an igniter. A liquid engine adds tanks, pumps and plumbing, and gets control in return.

5 Shedding weight: staging

Reaching low Earth orbit takes roughly 9 to 10 km/s of speed once gravity and drag losses are counted. We derive that in Chapters 2 and 3. For now, the problem is plain: an empty tank is dead weight that you are still accelerating. **Staging** solves it by dropping the empty parts, so the remaining engines push less mass. NASA files this under the structural system, because the structure is what gets discarded.

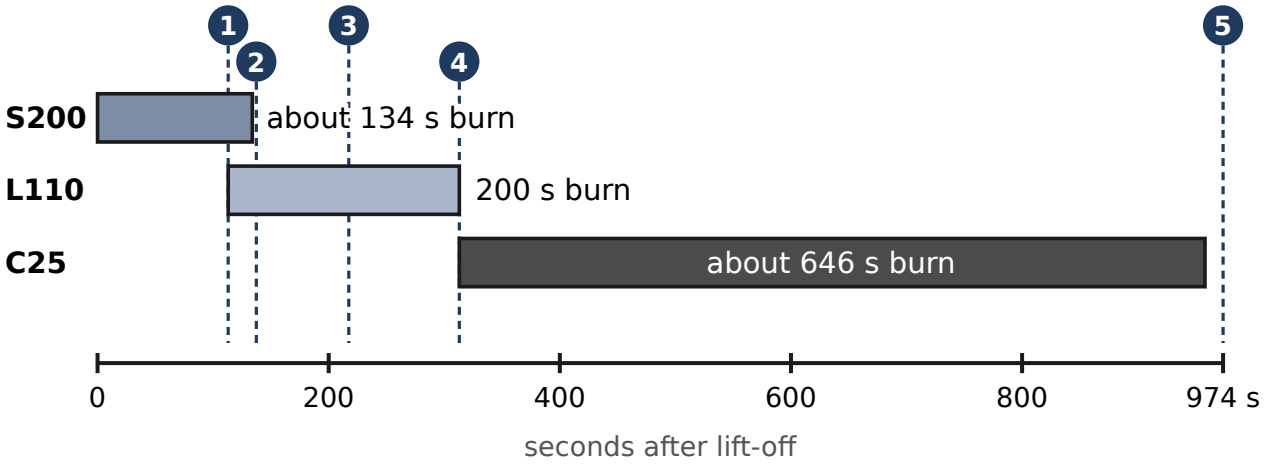


Figure 5. LVM3’s flight on a time axis, using ISRO’s nominal timings. Numbers refer to the events below.

1. **113 s:** the L110 core ignites at 41.5 km altitude while the boosters are still burning.
2. **137 s:** the two spent S200 boosters separate (they burned for about 134 s).
3. **217 s:** the payload fairing is jettisoned at 115 km, by then the air is thin enough that the shield is no longer needed.
4. **313 s:** the L110 burns out and separates; the C25 cryogenic stage ignites.
5. **974 s:** the satellite is injected into its 180 × 36,000 km transfer orbit.

6 Worked examples on LVM3

All figures are from ISRO’s LVM3 page: lift-off mass 640 t; two S200 boosters with about 205 t of propellant each; L110 with 115 t; C25 with 28 t and a dry mass of 4,310 kg; C25 thrust 200 kN (nominal); burn times 134 s, 200 s and about 646 s.

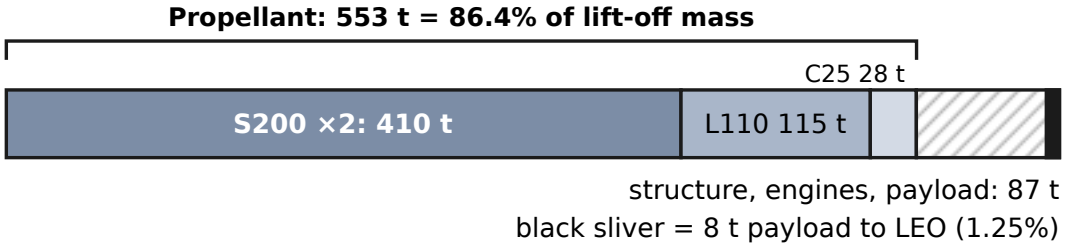
Example A: what must the boosters beat at lift-off?

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W = m × g = 640,000 kg × 9.81 m/s² = 6,278,400 N ≈ 6.28 MN
```

ISRO says LVM3 lifts off on the two S200 boosters alone; the L110 lights at 113 s. So the pair must produce more than 6.28 MN, which means **each S200 needs more than about 3.14 MN** just to hold the vehicle on the pad. With T/W = 1.5 (a what-if, not an ISRO figure), the acceleration would be 0.5 × 9.81 ≈ 4.9 m/s².

Example B: how much of the rocket is propellant?

```
Propellant = 2 × 205 + 115 + 28 = 553 t
Share      = 553 / 640 = 0.864  →  86.4%
Everything else = 640 − 553 = 87 t  (13.6%)
Payload to LEO = 8 t / 640 t = 1.25%
Payload to GTO = 4 t / 640 t = 0.625%
```



Of every 100 kg on the pad, about 86 kg is propellant that will be burnt, and about 1.25 kg arrives in low orbit as payload. A rocket is mostly inventory it consumes on the way up.

Example C: how fast does it burn propellant?

Mass flow rate is propellant mass divided by burn time. These are averages from nominal burn times; a solid motor's real flow rises and tails off.

```
S200 (each): 205,000 kg / 134 s  ≈ 1,530 kg/s
S200 (both): 2 × 1,530           ≈ 3,060 kg/s
L110       : 115,000 kg / 200 s  =   575 kg/s
C25        :  28,000 kg / 646 s  ≈    43 kg/s
```

At lift-off the two boosters burn about 3 tonnes of propellant every second, and each one burns about 35 times faster than the C25 does. Thrust equals mass flow times exhaust speed, so this is the number to hold onto for Chapter 1.

Example D: can an upper stage lift its own weight?

```
C25 mass (stage only) = 28,000 + 4,310 = 32,310 kg
Weight at Earth's surface g = 32,310 × 9.81 ≈ 317 kN
T/W = 200 kN / 317 kN ≈ 0.63
```

T/W is below 1, even before counting the satellite. **That is normal for an upper stage.** It ignites far above the ground and already moving fast, so it never has to hold itself up from rest. Only the stages that start on the ground need T/W above 1 at ignition.

7 Chapter summary

- A rocket works by **throwing its own exhaust backwards** (Newton's third law), so it needs no air and works in space.
- Four forces act on it: **weight, thrust, drag and lift**. On a rocket, thrust beats weight, drag is large, and lift is only a steering side force.
- **T/W above 1** is needed to leave the pad: $a = (T/W - 1) \times g$.
- Parts group into four systems: **structural, payload, guidance, propulsion**. Propulsion is most of the vehicle.
- **Solid** engines are simple and cannot be switched off; **liquid** engines are complex and controllable. LVM3 uses solid, liquid and cryogenic stages.
- About **86%** of LVM3's lift-off mass is propellant, and only about 1.25% reaches low orbit as payload. **Staging** drops empty weight to improve that.

Next, Chapter 1: Thrust and specific impulse. We turn "throw mass backwards" into a formula and find out why a cryogenic stage with a slow burn beats a booster that burns 35 times faster.

Practice problems

Use g = 9.81 m/s². Show your working on paper. Answers are on the last page.

P1. PSLV-XL lifts off at about 320 tonnes (a public-source figure). Find its weight at lift-off in newtons and in MN.

P2. A rocket has T/W = 1.2 at lift-off. Ignoring drag, what is its initial acceleration?

P3. What fraction of LVM3's total propellant sits in the two S200 boosters?

P4. LVM3 can carry 8,000 kg to low Earth orbit or 4,000 kg to geosynchronous transfer orbit. Express each as a percentage of lift-off mass. What does the ratio tell you about the cost of going higher?

P5. ISRO says the L110 ignites at 113 s and burns out at 313 s. Does that match its stated 200 s burn? Why can it ignite while the S200s are still burning?

P6. A toy balloon released with its neck open zooms around the room. Name the action, the reaction and the working fluid. Would it work inside a vacuum chamber?

Glossary

Thrust
The force an engine makes by throwing exhaust backwards. Measured in newtons (N) or kilonewtons (kN).

Weight
Mass × g, the pull of gravity on the vehicle.

Drag
Air resistance, opposite to the direction of flight.

Lift (on a rocket)
The aerodynamic side force from fins and body that helps stabilise and steer.

Centre of gravity (CG)
The balance point; weight acts through it.

Centre of pressure (CP)
The point where the air forces act. It should sit behind the CG.

Payload
What the rocket exists to deliver: a satellite, instruments or people.

Propellant
Fuel plus oxidiser, the stuff that is burnt.

Oxidiser
The oxygen source that lets fuel burn without outside air.

Working fluid
The gas that is accelerated to make thrust. For a rocket, its own exhaust.

Cryogenic
Using propellants (liquid oxygen, liquid hydrogen) kept at extremely low temperatures.

Staging
Dropping empty tanks and engines during flight so less mass is accelerated.

Gimbal
Swivelling the nozzle to steer by changing the thrust direction.

T/W
Thrust-to-weight ratio: thrust divided by weight.

Answer key

Fold this page back before you start the problems.

P1. W = 320,000 × 9.81 = 3,139,200 N ≈ **3.14 MN**. (Half of LVM3's 6.28 MN, because the vehicle is half as heavy.)

P2. a = (T/W − 1) × g = 0.2 × 9.81 ≈ **1.96 m/s²**, about 0.2 g.

P3. 410 t / 553 t ≈ 0.741, so **74.1%**. Three quarters of the propellant is in the boosters.

P4. LEO: 8,000 / 640,000 = **1.25%**. GTO: 4,000 / 640,000 = **0.625%**. Going higher halves the payload fraction: each step up the energy ladder costs a lot of mass. (Chapter 2 shows exactly why.)

P5. 313 − 113 = **200 s**, which matches. Each stage has its own engines and its own propellant, and a liquid engine can be started on command. So ISRO lights the core at 41.5 km altitude while the boosters are still burning, and drops the boosters later.

P6. Action: the balloon throws air out of its neck. Reaction: the balloon is pushed the other way. Working fluid: the air that was inside the balloon. **Yes, it works in a vacuum**, because it carries its own working fluid. This is the same reason a rocket works in space.

Sources

NASA Glenn Research Center, Beginner's Guide to Rockets / Aeronautics (www1.grc.nasa.gov/beginners-guide-to-aeronautics): Newton's Laws of Motion; Four Forces on a Rocket; Rocket Parts; Structural System; Propulsion System. The explanations are rewritten in my own words and all diagrams are my own.

ISRO, LVM3 launch vehicle page (isro.gov.in/GSLVmk3_CON.html), checked 9 November 2026. All LVM3 figures are nominal values published by ISRO. ISRO's page does not give lift-off thrust, so none is used here.

PSLV-XL lift-off mass (about 320 t) comes from public sources, not ISRO's page, and is used only in practice problem P1.